Colab Overview

Hands-On Financial Trading with Python, Second Edition

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Why we use Google Colab



Nothing to install

It runs in your web browser.



Free to start

You do not need paid software to follow along.



Common packages are preinstalled

pandas, NumPy and Matplotlib are ready to import.



Same setup on any computer

Windows, macOS or Linux, it looks the same.

A quick tour: the menus



File

- Open notebook
- Save a copy in Drive
- Download .ipynb

Runtime

- Run all
- Restart session
- Change runtime type

Tools

- Settings
- Command palette
- Keyboard shortcuts



Connect starts your session: the computer in the cloud that runs your code.

Opening a notebook from the repo

1

Click File, then Upload notebook

This opens a file picker.

2

Pick the notebook from your cloned folder

The repo folder you cloned in Video 1.

3

Colab opens it, ready to run

Click Connect, then run the cells from the top.

One upload per notebook. After that, you work in Colab.

When I update a notebook

Fetch and Pull in GitHub Desktop, then upload the new version. Rename your edited copy first if you want to keep it.

A quick tour: the left pane



Table of contents — Jump between sections using the notebook's headings.

Find and replace — Search the whole notebook.

Variables — See what is in memory right now.

Secrets — A private place to keep API keys.

Files — The session's folder. Files here disappear when the session ends.

The icons can change slightly as Colab updates.

Try it: a 30-second demo

Cell 1

```
!pip install -q yfinance
```

Cell 2

```
import pandas as pd
import numpy as np
import yfinance as yf
print(pd.__version__, np.__version__)
```

Cell 3

```
spy = yf.download("SPY", period="1y",
                  auto_adjust=True, progress=False)
spy["Close"].plot(title="SPY, last year");
```

Watch for

- The install cell runs first.
- The version numbers print. Remember them.
- A chart appears under the cell.
- Run a cell with Play or Shift+Enter.



Screen recording follows.

What can go wrong



Why it happens

- Google updates the preinstalled packages, and sometimes Python itself
- Functions get renamed, changed or removed
- Packages can stop agreeing on versions
- Installs vanish when your session resets



What you see

- An ImportError or AttributeError
- Red dependency warnings after an install
- Results that differ slightly from the book

Example: pandas renamed the monthly resample code from M to ME.

Fix 1: pin the version

Pin when

- A notebook breaks after an update
- You need to reproduce the book's numbers exactly
- Code runs on a schedule, like Chapters 17 and 18

Do not pin when

- You are only exploring and everything runs
- A preinstalled package already works
- It is NumPy or pandas: pinning can force restarts or clash with other packages

```
!pip install package==1.2.3
```

Pin only the package that breaks.

Fix 2: ask an AI assistant

1

Copy the error

The full message, and the cell that failed.

2

Share your versions

Print them, as in the demo, and paste them in.

3

Ask for the smallest fix

A one-line change, not a rewrite.

4

Check against the book

Rerun and compare. If the numbers moved, the fix changed too much.



Also check the repo for a fixed notebook: Fetch origin, then Pull origin.

Colab in four points



No setup: it runs in your browser



Know the menus and the left pane



If a package breaks, pin only that package



Or ask an AI assistant for the smallest fix

Next: open a notebook and start Chapter 1